

Problem 1 (30 marks in total)

Consider a system represented by the linear differential equation:

$$y^{(2)} + 5y^{(1)} + 6y = 5u^{(1)} + 4u.$$

with input u and output y , and where $y^{(n)}$ and $u^{(n)}$ denote the n -th derivative of y and u with respect to t , respectively. Answer the following questions:

- (a) **(5 marks)** Determine the observer canonical form for the system.
- (b) **(10 marks)** Sketch a block diagram of the observer canonical form for this system.
- (c) **(15 marks)** Using the observer canonical form, apply Ackermann's formula to calculate a linear state-feedback law $u = -Kx$ such that the closed-loop matrix $A - BK$ has eigenvalues at $-3 \pm 4j$. Show all the steps, including the calculation of $R(A)$.

Problem 2 (35 marks in total)

Consider the state space system

$$S = \left(\begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, [0.5 \quad -0.5], 0 \right)$$

Answer the following questions:

- (a) **(5 points)** Compute the system's transfer function.
- (b) **(5 marks)** Is the system BIBO stable?
- (c) **(5 marks)** Find the transformation matrix T that transforms the system into modal canonical form.
- (d) **(10 marks)** Using the transformation T for sub-problem (c), compute the modal canonical form.
- (e) **(10 marks)** Using the modal canonical form from sub-problem (d), calculate the solution $x(t)$ and $y(t)$ of the system for any input u when the initial condition is $x_0 = \begin{bmatrix} 0 & 1 \end{bmatrix}^T$.

Problem 3 (35 marks in total)

- (a) Consider the nonlinear system on $S = \mathbb{R}^2$ described by the ODE:

$$\begin{aligned}\dot{x} &= -x + 3(y - 1) + x^3 \\ \dot{y} &= -3x - (y - 1).\end{aligned}$$

Answer the following questions:

- (i) **(5 marks)** The system has three equilibria. One is at $x^* = (0, 1)$, where are the other two? Show your calculations.

- (ii) **(10 marks)** Consider the candidate Lyapunov function $V(x, y) = x^2 + (y - 1)^2$. Compute $\dot{V}(x, y)$ and show whether it is lnsd or lnd around $(0, 1)$.

(b) Consider the nonlinear controlled ODE on $S = \mathbb{R}^2$ given by:

$$\begin{aligned}\dot{x}_1 &= x_2 - (x_1 - x_2)x_2^3 \\ \dot{x}_2 &= -x_1 + (x_1 - x_2)x_2^3 - (x_1 - x_2)^2 u,\end{aligned}$$

where u is a scalar control signal. Answer the following questions:

- (i) **(10 marks)** Consider the candidate Lyapunov function $V(x_1, x_2) = x_1^2 + x_2^2$. Calculate $\dot{V}(x_1, x_2)$ and then determine a control law u such that

$$\dot{V}(x_1, x_2) = -2\kappa(x_1 - x_2)^2 x_2^2$$

for some constant $\kappa > 0$.

- (ii) **(10 marks)** Under the control law in sub-problem (i) above, compute the largest positively invariant set M in the set $I = \{(x_1, x_2) \in \Omega_{r,c} \mid \dot{V}(x_1, x_2) = 0\}$ for any $r, c > 0$ such that $\Omega_{r,c} = \{x \in \mathbb{R}^2 \mid \|x\| \leq r, V(x) \leq c\}$ is positively invariant (note that $\Omega_{r,c}$ contains the origin).